UniformEvalBench: A Multi-Axis Benchmark and Clinically Grounded Downstream Task for EEG Foundation Models

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Abstract

Current evaluation of EEG Foundation Models (EEG-FMs) relies primarily on full fine-tuning accuracy across curated downstream datasets. This practice conflates three distinct properties: the quality of the pretrained representation, the capacity of a trainable task head to reorganize that representation, and dataset-specific inductive biases of the backbone architecture. We introduce UNIFORMEVALBENCH, a multi-axis benchmark that evaluates EEG-FMs along three complementary dimensions: *representation quality* via kNN@20, a hyperparameter-free metric requiring no learned probe, together with formally defined adaptation metrics (ADAPTGAP, NormAdaptGain); *robustness* via Channel Degradation Resilience (CDR), a chance-corrected score measuring how far above chance a model remains under test-only channel dropout simulating electrode failure; and *neurophysiological grounding* via Spectral Prior Alignment (SPA), a task-agnostic metric measuring whether frozen embeddings linearly encode canonical EEG frequency bands. A composite leaderboard (UEB-GENERAL) aggregates the three axes via a geometric mean that prevents single-axis dominance. The benchmark pre-registers three confirmatory claims with quantitative success criteria and separates them from exploratory analyses. We evaluate 6 EEG-FMs and 1 general time-series FM across 7 datasets spanning motor imagery, sleep staging, seizure detection, neurodegenerative diagnosis, cognitive workload, and a *novel clinically grounded downstream task*: discriminating epilepsy from its clinical mimickers, for which we publicly release a new 80-subject dataset (50 epileptic, 30 mimickers including PNES, syncope, and dystonia) recorded with standard 19-channel 10–20 montage EEG. This discrimination problem is of high clinical significance—misclassification leads to years of inappropriate treatment—yet has never been posed as a foundation model benchmark. Our results demonstrate that frozen and fine-tuned evaluations rank models differently (mean Kendall's $\tau = 0.291$ across 7 datasets), that the strongest frozen-representation model does not lead on robustness, and that spectral encoding is a necessary condition for downstream performance (FBP–SPA $\rho = +0.90$). Code, benchmark harness, and data are publicly released at `https://github.com/ANONYMOUS/UniformEvalBench`.

1 Introduction

EEG Foundation Models (EEG-FMs) such as BIOT [Yang et al., 2023], LaBraM [Jiang et al., 2024], EEGPT [Wang et al., 2024a], CBraMod [Wang et al., 2024b], CSBrain, and REVE span diverse architectures: patch-based vision transformers, criss-cross attention networks, and deep transformers with Fourier positional encoding. Despite this diversity, the field evaluates them through a near-uniform protocol: fine-tune on a downstream dataset, report balanced accuracy. This

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Table 1: EEG evaluation benchmark comparison. UNIFORMEVALBENCH is the first to jointly address frozen-backbone probing, robustness, neurophysiological grounding, pre-registration, and a novel clinical task.

Benchmark	FBP	Robust.	Neuro Gr.	Pre-Reg.	Novel Task
MOABB	–	–	–	–	–
BEETL	–	–	–	–	–
AdaBrain-Bench	–	–	–	–	–
EEG-FM-Bench	Partial	–	–	–	–
UNIFORMEVALBENCH	✓	✓	✓	✓	✓

practice has three systematic shortcomings. First, **fine-tuning conflates representation quality with adaptation plasticity**: a sufficiently expressive task head can achieve strong performance on a mediocre backbone, so FT accuracy cannot attribute performance to the pretrained encoder. Second, **clean-data accuracy overestimates deployment reliability**: deployed EEG systems see electrode failures, motion artifacts, and non-stationary noise that curated benchmarks remove. Third, **accuracy does not certify neurophysiological validity**: a model exploiting EMG bleedthrough in motor imagery data can score well without encoding neurophysiologically plausible features.

EEG-FM-Bench [Xiong et al., 2026] documented these issues with diagnostic analyses (gradient conflicts, CKA/RSA, scaling laws) but retained FT accuracy as its primary ranking criterion. Existing benchmarks also lack a clinically challenging downstream task: they evaluate on paradigms where EEG signals are either grossly abnormal (seizure detection, sleep staging) or paradigm-locked (motor imagery). They do not test whether EEG-FMs encode the subtle interictal signatures that distinguish conditions presenting identically to the clinician but differing in etiology and treatment.

Epilepsy vs. its clinical mimickers is the canonical instance of this problem. Non-epileptic conditions—PNES, syncope, dissociative reactions, movement disorders, metabolic encephalopathy—produce presentations frequently indistinguishable from epileptic seizures. Misdiagnosis has severe consequences: 20–30% of tertiary epilepsy center referrals carry an incorrect initial diagnosis [Benbadis & Allen Hauser, 2000], and the mean diagnostic delay for PNES exceeds 7 years [Reuber et al., 2002]. Routine interictal EEG is normal in up to 50% of confirmed epilepsy patients [Smith, 2005], making this discrimination a demanding benchmark for whether EEG-FM representations capture subtle, clinically meaningful signal structure.

Contributions. We introduce UNIFORMEVALBENCH with five contributions: (1) a hyperparameter-free primary metric (kNN@20) for frozen-encoder quality, following DINOv2 [Oquab et al., 2023]; (2) a chance-corrected robustness metric (CDR) that measures absolute above-chance performance under electrode failure, unlike the retention ratio which rewards trivially stable near-chance models; (3) Spectral Prior Alignment (SPA), a prior-free grounding metric; (4) a novel 80-subject epilepsy-vs-mimickers dataset and clinical downstream task; and (5) a composite leaderboard (UEB-GENERAL) and adapter-based evaluation harness (`ModelAdapter` ABC) for extensibility. Three confirmatory claims are pre-registered with quantitative thresholds and fallback narratives. Code and data: `https://github.com/ANONYMOUS/UniformEvalBench`.

2 Related Work

MOABB [Jayaram & Barachant, 2018] standardizes BCI evaluation but predates foundation models. **BEETL** [Wei et al., 2022] targets transfer learning for motor imagery but does not evaluate frozen representations. **AdaBrain-Bench** [Chen et al., 2025] evaluates adaptation methods but does not measure representation quality, robustness, or grounding. **EEG-FM-Bench** [Xiong et al., 2026] provides comprehensive diagnostic analyses but retains FT accuracy as primary. UNIFORMEVALBENCH provides an independent adapter-based evaluation harness: the `ModelAdapter` ABC does not depend on any prior benchmark's internals, and any external EEG-FM can be evaluated by implementing three methods (`encode`, `freeze_encoder`, `unfreeze_encoder`). Table 1 summarizes the comparison.

Table 2: Dataset pool. D7 (Epilepsy Mimickers) is a novel public dataset contributed by this work.

#	Dataset	Paradigm	Cls.	Subj.	Dur.	Role
D1	BCIC-IV-2a	Motor imagery	4	9	4 s	Standard BCI
D2	HMC	Sleep staging	5	151	30 s	Clinical sleep
D3	ADFTD	Dementia	3	88	10 s	Neurodegenerative
D4	Motor-MV	Motor imagery	4	–	4 s	Multi-view motor
D5	Siena Scalp	Seizure det.	2	14	var.	Clinical binary
D6	Workload	Cogn. workload	2	–	var.	Cognitive state
D7	**Epil. Mim.**	**Epil. vs. Mim.**	**2**	**80**	**10 s**	**Novel clinical**

3 Benchmark Design

Three principles govern the design: (1) *Separation of concerns*: each axis measures a distinct property; if two axes correlate perfectly, one is redundant. (2) *Realistic degradation*: robustness uses electrode-failure simulation, not Gaussian noise. (3) *Validity checks*: novel metrics must pass empirical validation against independent outcomes.

3.1 Model Pool

We evaluate **6 EEG-FMs** (BIOT, LaBraM, EEGPT, CBraMod, CSBrain, REVE) plus **MOMENT** [Goswami et al., 2024] as a general time-series FM reference. Each model is integrated via a `ModelAdapter` subclass wrapping model-specific weight loading and forward pass; the benchmark harness calls only adapter methods, ensuring model-agnostic evaluation. All models expose a pooled feature vector from their final encoder block; L2-normalization is applied before probing. Full architectural details are in Appendix A.

3.2 Dataset Pool and D7: Epilepsy vs. Clinical Mimickers

The benchmark uses **7 datasets** spanning major EEG paradigms (Table 2). Datasets D1–D6 are established; D7 is a novel contribution. All datasets are resampled to 200 Hz, bandpass filtered (0.5–75 Hz), notch filtered at 50/60 Hz, and re-referenced to common average. Trial durations are paradigm-specific to preserve task-defining signal structures.

D7: Epilepsy Mimickers. The dataset comprises **80 subjects**: 50 epileptic (E001–E050) spanning generalized, focal, and juvenile myoclonic subtypes, and 30 mimickers (M001–M030) encompassing PNES, syncope, dissociative reactions, movement disorders, vertigo, and metabolic causes. All recordings use 19-channel 10–20 montage at 125 Hz with linked-ear reference, segmented into non-overlapping 10-second epochs with the whole-recording binary label. Splits are deterministic and subject-level: 64 train / 8 validation / 8 test subjects. A repeated 5-fold stratified subject-level cross-validation is additionally reported as a sensitivity analysis. Full subject-level metadata are in Appendix G. This dataset is publicly released alongside the benchmark.

Clinical significance. Epilepsy is among the most common serious neurological disorders, affecting approximately 50 million people worldwide [World Health Organization, 2023]. The consequences of misdiagnosis are severe and asymmetric: a patient with PNES misdiagnosed as epileptic may receive anti-epileptic drugs for years without benefit and with substantial side effects; conversely, a patient with true epilepsy misdiagnosed as having a functional disorder is exposed to uncontrolled seizure risk. Routine interictal scalp EEG is the first-line investigation, but its discriminative power for this task is limited: interictal EEGs are normal in up to 50% of confirmed epilepsy patients [Smith, 2005]. Unlike seizure *detection*, which asks "is there an obvious paroxysmal event?", the mimicker task asks "does this EEG come from a brain with an epileptic network, even in the absence of overt ictal activity?" This requires the model to encode subtle interictal signatures: low-amplitude epileptiform discharges, focal slowing, altered background rhythms, and disrupted spectral architecture. If EEG-FMs have truly learned the deep structural regularities that define neural signal, they should distinguish these conditions from their frozen representations alone.

4 Evaluation Methodology

4.1 Axis A: Representation Quality

kNN@20 (primary). Any trained probe introduces hyperparameter choices that can systematically favour one model's geometry over another's. kNN@20 eliminates this: it has exactly one hyperparameter (k, a count), learns nothing, and directly measures whether same-class samples cluster in embedding space—the evaluation strategy adopted by DINOv2 [Oquab et al., 2023]. We sweep $k \in \{5, 10, 20, 50\}$ and report k=20; k-sensitivity is reported in Appendix E. For cross-dataset comparability, the Axis A score applies chance correction:

$$A_m = \frac{1}{D} \sum_d \max\left(0,\ \frac{\text{BA}^{k\text{NN}}_{m,d} - c_d}{1 - c_d}\right), \quad c_d = \frac{1}{K_d} \tag{1}$$

FBP (secondary). Encoder frozen; classifier head (Linear→ReLU→Dropout→Linear) trained with AdamW, LR $= 5\times10^{-4}$, 30 epochs. Validated against kNN@20 via Kendall τ (see §5.1).

Full fine-tuning (FT). Same head, encoder unfrozen; encoder LR $= 0.1\times$ head LR, matching each model's published recipe (per-model details in Appendix A).

Adaptation metrics.

$$\text{AdaptGap}(m,t) = \text{FT}(m,t) - \text{FBP}(m,t) \tag{2}$$

$$\text{NormAdaptGain}(m,t) = \frac{\text{FT}(m,t) - \text{FBP}(m,t)}{1 - \text{FBP}(m,t)} \tag{3}$$

NormAdaptGain $\in (-\infty, 1]$; negative values indicate FT *harms* performance.

4.2 Axis B: Robustness Under Channel Dropout

The retention ratio ($\text{BA}^{\text{corrupt}}/\text{BA}^{\text{clean}}$, averaged over datasets and dropout levels) answers "how stable is the model relative to itself?" but not "how useful is the model when channels are missing?" A model at clean BA $= 0.52$ and corrupt BA $= 0.50$ achieves Ret.Ratio $= 0.96$ yet remains useless. We define CDR as the chance-corrected absolute performance under corruption:

$$\boxed{\text{CDR}_m = \frac{1}{|D||P|} \sum_{d,p} \max\left(0,\ \frac{\text{BA}^{\text{corrupt}}_{m,d,p} - c_d}{1 - c_d}\right)} \tag{4}$$

where $P = \{0.1, 0.25, 0.4\}$ are random-dropout severity levels applied *pre-adapter, test-only* (train-clean, test-corrupt). Model selection uses clean validation data. The retention ratio is retained as a diagnostic column.

4.3 Axis C: Spectral Prior Alignment

SPA measures whether frozen embeddings linearly encode the five canonical EEG frequency bands:

$$\text{SPA}(m,t) = \overline{R^2}\Big(\mathbf{z}_{m,t} \xrightarrow{\text{Ridge}} [P_\delta,\ P_\theta,\ P_\alpha,\ P_\beta,\ P_\gamma]\Big) \tag{5}$$

where $\mathbf{z}_{m,t}$ is the frozen pooled embedding, P_b is relative band power (Welch PSD, averaged across channels, normalized to sum to 1), and Ridge regression ($\alpha \in [10^{-3}, 10^4]$, 5-fold CV) maps embeddings to band powers. SPA $\in [0, 1]$; higher is better. BIOT is excluded from SPA because its STFT tokenizer trivially encodes band powers by construction. EEG neuroscience is organized around spectral content: sleep staging is defined by δ waves and spindles [Berry et al., 2012], motor imagery by μ/β event-related desynchronization [Pfurtscheller & Lopes da Silva, 1999], and cognitive load by frontal midline θ [Onton et al., 2005]. If a pretrained EEG-FM does not linearly encode these spectral features, its embedding geometry does not reflect the primary organizational principle of the field. SPA tests this directly without requiring any spatial prior or downstream task labels.

Table 3: kNN@20 leaderboard, ranked by chance-corrected A_m. Global k-sensitivity = 0.029 (Appendix E).

Rank	Model	A_m	kNN raw	FBP mac.	Δ
1	EEGPT	**0.229**	0.502	0.571	-0.069
2	BIOT	0.196	0.484	0.533	-0.048
3	CSBrain	0.163	0.462	0.471	-0.009
4	REVE	0.136	0.453	0.585	-0.131
5	CBraMod	0.111	0.421	0.461	-0.040
6	LaBraM	0.097	0.421	0.442	-0.021

4.4 Composite: UEB-GENERAL

Axes A and C share variance (FBP–SPA $\rho = +0.90$); summing them would double-count representation quality. We define RepCore $= 0.75\,A_m + 0.25\,C_m$ (where $C_m = \frac{1}{D}\sum_d \text{SPA}_{m,d}$), and the composite ranking via a geometric mean:

$$\boxed{\text{UEB-General}_m = (\text{RepCore}_m + \varepsilon)^{0.6}\,(\text{CDR}_m + \varepsilon)^{0.4} - \varepsilon, \quad \varepsilon = 10^{-4}} \tag{6}$$

The 0.6/0.4 exponents match the RepCore/CDR importance ratio. A model with RepCore $= 0.90$ and CDR $= 0.01$ scores ≈ 0.09; an arithmetic mean would yield 0.544, incorrectly suggesting near-best performance. The geometric mean structure ensures a model cannot achieve top ranking by excelling on one dimension while collapsing on another. Note that A_m takes values in $[0, \sim 0.25]$ for the current model pool while C_m takes values in $[0.57, 0.89]$, so the 25%-weighted C_m contributes comparably to A_m in absolute magnitude. This is intentional: C_m serves as a spectral grounding floor check; A_m provides cross-dataset discrimination. Models missing any axis appear in a partial leaderboard.

5 Results

We evaluated 7 models across 7 datasets $\times$ 3 seeds. Axis A: 240 FBP+FT runs and 141 kNN runs completed. Axis B: 378 corrupted-test runs. Axis C: 105 SPA runs (all complete). MOMENT was excluded from D7 (memory constraints) and from D4/D5 FT (OOM).

5.1 Axis A: Representation Quality

kNN@20 leaderboard. Table 3 presents the primary representation quality ranking. EEGPT leads ($A_m = 0.229$); REVE has the largest kNN–FBP gap (-0.131), indicating its probe-inflated FBP does not reflect true embedding separability. Mean $\tau_{k\text{NN,FBP}} = 0.519 < 0.7$ across 7 datasets: FBP is **not** a certified proxy for kNN@20, confirming kNN@20 as the sole primary metric (full proxy analysis in Appendix E).

FBP–FT ranking divergence (Claim 1). Per-dataset Kendall's τ between FBP and FT rankings yields mean $\tau = 0.291$ with bootstrap 95% CI $[0.03, 0.56]$, well below the pre-registered 0.7 threshold. **Claim 1 is confirmed.** The best-model identity flips on 5 of 7 datasets (Table 4). Per-dataset tables with kNN@20, FBP, FT, ADAPTGAP, and NormAdaptGain are in Appendix B.

Fine-tuning harms on D7 and ADFTD. On Epilepsy Mimickers, 5 of 6 models show negative ADAPTGAP (Table 5). EEGPT achieves 0.877 FBP—the highest in the entire benchmark—yet collapses to 0.676 under FT (AdaptGap $= -0.202$). The same pattern appears on ADFTD (5/7 negative). Full ADAPTGAP span across the benchmark: -0.103 (BIOT on ADFTD) to $+0.289$ (REVE on Motor MV), a 39-point range exceeding cross-model clean-accuracy variance.

Per-model profiles. **EEGPT** dominates both kNN@20 and FBP (rank #1 on 4/6 core datasets) but drops under FT on those same datasets, indicating its representations are near their effective ceiling. **REVE** leads FT (rank #1 on 3/6) despite ranking 4th on kNN@20: its frozen embedding geometry is not naturally class-clustered but responds well to adaptation. **BIOT** ranks 2nd on kNN@20 vs 4th on FBP—its STFT tokenization produces well-clustered embeddings with little trained-probe benefit. **CBraMod** exhibits the largest FBP-to-FT headroom consistently: frozen features are

Table 4: FBP vs. FT ranking agreement across all 7 datasets. Mean $\tau = 0.291 < 0.7$ (Claim 1 confirmed).

Dataset	τ	ρ	FBP #1	FT #1
BCIC-IV-2a	+0.048	0.00	EEGPT	MOMENT
HMC	+0.238	+0.43	EEGPT	REVE
Siena Scalp	+0.276	+0.43	EEGPT	CBraMod
Motor MV	+0.467	+0.60	EEGPT	REVE
ADFTD	+0.619	+0.75	REVE	REVE
Workload	+0.720	+0.82	REVE	REVE
Epil. Mim.	-0.333	-0.54	EEGPT	REVE
Mean	**+0.291**	**+0.36**		

Table 5: Axis A on D7 (Epilepsy Mimickers). 5/6 models: FT *harms* frozen features.

Model	kNN	FBP	FT	AdGap	NAG
EEGPT	0.794	**0.877**	0.676	-0.202	-1.644
CSBrain	0.730	0.827	0.662	-0.164	-0.948
REVE	0.716	0.819	0.767	-0.052	-0.285
BIOT	0.696	0.765	0.716	-0.048	-0.205
LaBraM	0.723	0.751	0.743	-0.008	-0.034
CBraMod	0.723	0.708	0.729	$+0.022$	$+0.074$

often near chance, but FT brings it to top-of-pack (siena_scalp FBP 0.619 → FT 0.849). **CSBrain** stalls at exactly 0.500 FBP on siena_scalp and workload (the trained probe produces constant-class predictions), yet FT recovers dramatically (+0.278 gap on Motor MV). **LaBraM** is consistently bottom-ranked on both FBP and FT; its VQ-NSP representations may not be well-matched to these downstream paradigms. **No Pareto-optimal model exists**: the best kNN model (EEGPT) is rarely the best FT model (REVE).

5.2 Axis B: Robustness

Table 6 presents the CDR leaderboard for the 5 verified EEG-FMs. BIOT is excluded from Axis B: its STFT tokenizer applies a per-channel frequency-domain transform before the dropout hook can act on the raw signal, producing a pipeline-artifact CDR that is not comparable to the other models. Among verified models, EEGPT leads (0.223). The CDR–Ret.Ratio divergence is exactly as predicted: CSBrain achieves the highest Ret.Ratio (0.985) but only 3rd-highest CDR (0.166), because its near-chance clean performance is trivially dropout-stable. Clean performance and CDR are weakly positively correlated ($\tau \approx +0.47$): stronger models degrade more in absolute terms but remain more useful.

Siena Scalp is the hardest dataset for robustness: four models collapse to exactly chance (0.500) at $p \geq 0.25$. Only EEGPT and BIOT remain above chance at $p = 0.4$. Binary seizure detection relies heavily on high-gamma and sharp-wave features that are spatially concentrated; losing 10+ channels destroys the relevant signal. D7 (Epilepsy Mimickers) is the most discriminating dataset: cross-model spread at $p = 0.4$ (0.500–0.735) is larger than any other dataset, and EEGPT retains the highest per-dataset CDR (0.635). On ADFTD, REVE's accuracy *improves* under dropout (clean 0.464 → 0.567 at $p = 0.25$): REVE's channel-specific fine-grained features introduce noise on the 3-class task, and channel zeroing acts as regularization that improves cross-subject generalization. Per-dataset CDR and degradation curves are in Appendix C.

5.3 Axis C: Spectral Grounding

REVE leads SPA (0.893) with uniformly high per-band R^2 (0.832–0.906); EEGPT is second (0.863). LaBraM is last (0.577) with θ-band R^2 of 0.496, mechanistically explaining its near-chance FBP on frontal-θ-dependent tasks. FBP–SPA $\rho = +0.90$ confirms spectral encoding underlies representation quality. SPA–Ret.Ratio $\rho = -0.70$ reveals a fundamental tension: channel-robust models achieve invariance through spectral compression. D7 has the second-highest SPA across all models (0.83–

Table 6: CDR (primary) and Ret.Ratio (diagnostic). 5 verified EEG-FMs × 7 datasets × 3 seeds × 3 dropout rates. BIOT excluded: STFT tokenizer bypasses pre-adapter channel dropout (pipeline artifact).

Rank	Model	CDR	Ret.Ratio	n
1	EEGPT	**0.223**	0.893	21
2	REVE	0.171	0.880	21
3	CSBrain	0.166	**0.985**	21
4	LaBraM	0.127	0.975	21
5	CBraMod	0.112	0.975	21

Table 7: SPA leaderboard (5 models × 7 datasets × 3 seeds = 105 runs). BIOT excluded.

Model	SPA	δ	θ	α	β	γ	Dim
REVE	**.893**	.906	.832	.895	.904	.899	512
EEGPT	.863	.868	.843	.894	.882	.796	256
CSBrain	.763	.763	.673	.736	.781	.760	200
CBraMod	.715	.694	.628	.695	.700	.739	200
LaBraM	.577	.637	**.496**	.556	.593	.538	256

0.92), confirming that the epileptic/mimicker distinction maps onto linearly accessible spectral differences. EEGPT is the most consistent model across all seven datasets: its SPA range is only 0.825–0.883, confirming paradigm-agnostic spectral encoding. CBraMod shows the strongest task dependence, dropping from HMC SPA = 0.913 to BCIC-2A SPA = 0.597—its patch-based tokenizer captures local high-frequency content differently from slow oscillations spanning multiple patches. Per-band profiles and per-dataset heatmaps are in Appendix D.

5.4 Composite: UEB-GENERAL

Table 8 presents the final ranking. EEGPT leads (UEB-GENERAL $= 0.310$), REVE second (0.251), CSBrain third (0.242). No model simultaneously achieves high A_m, high CDR, and high C_m. REVE ranks 2nd despite 4th on A_m—its high SPA (0.892) elevates RepCore, and its CDR (0.171) holds. High Ret.Ratio (≈0.97–0.99 for CSBrain, CBraMod, LaBraM) does not translate to high CDR: retaining near-chance performance under corruption is not meaningful robustness, and the geometric structure of UEB-GENERAL correctly ranks absolute CDR above the retention ratio.

BIOT is excluded from both Axis B (STFT tokenizer bypasses the channel-dropout hook, producing pipeline-artifact CDR) and Axis C (STFT tokenizer trivially encodes band powers); it appears in the partial leaderboard with $A_m = 0.196$ only. MOMENT is excluded from CDR due to OOM on the robustness sweep and appears in the partial leaderboard with strong SPA but indeterminate robustness.

6 Statistical Framework and Pre-Registered Claims

All experiments use 3 random seeds (42, 123, 7). Three confirmatory claims are subject to Bonferroni correction ($\alpha = 0.05/3 \approx 0.0167$); all other analyses are exploratory. Bootstrap 95% CIs (10,000 iterations) are reported around all aggregate statistics. Per-subject accuracy is the unit of analysis for paired tests where subject-level splits are available.

Claim 1 (FBP $\neq$ FT ranking): mean $\tau < 0.7$. **Confirmed** ($\tau = 0.291$, CI $[0.03, 0.56]$ across 7 datasets). The result does not depend on any single dataset: removing the most-disagreeing dataset (Epil. Mim., $\tau = -0.333$) gives mean $\tau = 0.395$, still well below threshold. **Claim 2** (chance correction re-ranks robustness; CDR is not redundant with Ret.Ratio): $|\tau_{\text{Kendall}}(\text{CDR}, \text{Ret.Ratio})| \geq 0.4$. **Confirmed** ($\tau \approx -0.80$): the top-3 models by Ret.Ratio (CSBrain, LaBraM, CBraMod) are bottom-3 by CDR, because near-chance models trivially retain near-chance performance under corruption—demonstrating that chance correction, not retention, captures meaningful robustness. **Claim 3** (SPA exposes model-specific spectral failure): per-band R^2 gap ≥ 0.10 between consecutive models. **Confirmed** (LaBraM θ $R^2 = 0.496$ vs CBraMod 0.628, gap = 0.132). LaBraM's overall

Table 8: UEB-GENERAL leaderboard. BIOT lacks both CDR (STFT bypasses dropout hook) and C_m (STFT trivially encodes band powers); MOMENT lacks CDR (OOM).

Rk	Model	A_m	CDR	R.R.	C_m	**UEB-G**
1	EEGPT	**.229**	**.223**	.893	.860	**.310**
2	REVE	.136	.171	.880	**.892**	.251
3	CSBrain	.163	.166	.985	.757	.242
4	CBraMod	.111	.112	.975	.711	.186
5	LaBraM	.097	.127	.975	.576	.175
–	BIOT	.196	N/A	N/A	N/A	partial
–	MOMENT	.091	N/A	–	.623	partial

SPA rank (5) is one position below its FBP rank (4); the confirmatory test is the band-specific R^2 gap, not aggregate rank displacement.

Model-selection sensitivity. Across the 6 core datasets for which both selectors are recorded, switching from last-epoch to best-validation-epoch selection changes mean τ from 0.395 to 0.495 (+0.10), in the direction of *more* agreement between FBP and FT—both selectors confirm Claim 1. We report last-epoch as primary, consistent with the EEG-FM literature and chosen before the confirmatory sweep (full sensitivity analysis in Appendix F).

Fallback narratives. If Claim 1 failed, FBP would be reframed as a $10\times$ cheaper equivalent to FT. If Claim 3 failed, SPA becomes a mechanistic explanation for FBP (the $\rho = 0.90$ correlation is itself a finding). All fallbacks are pre-specified.

7 Discussion and Limitations

No Pareto-optimal model. EEGPT leads on kNN@20 and CDR but ranks second on SPA. REVE leads on SPA but ranks fourth on kNN@20. LaBraM is the most dropout-stable (highest Ret.Ratio) but spectrally impoverished. This multi-dimensional landscape is invisible under FT-only evaluation. The three-axis geometry identifies distinct *model archetypes*: REVE is spectrally precise and accurate but channel-specific (FBP #1, SPA #1, Ret.Ratio 5/6); EEGPT is the most consistent spectral encoder with the strongest absolute robustness; CSBrain is the closest to balanced across all three axes; CBraMod is spectrally inconsistent (HMC SPA = 0.913 but BCIC-2A SPA = 0.597); and LaBraM is channel-robust but spectrally impoverished, occupying the (low-SPA, high-Ret.Ratio) corner that reflects the structural cost of discrete tokenization.

The epilepsy-mimickers diagnostic. D7 amplifies the FT-harms pattern that also appears on ADFTD: 5/6 models on D7 and 5/7 models on ADFTD show negative ADAPTGAP. D7 is not unique in producing this pattern, but it produces it at larger magnitude (mean ADAPTGAP -0.075 on D7 vs -0.026 on matched-model ADFTD), making it the more diagnostic of the two. EEGPT's frozen features achieve 87.7% balanced accuracy on D7—the highest frozen score in the benchmark—yet FT collapses this by 20 points. CBraMod is the sole exception on D7 (+0.022), likely because its cross-attention head benefits most from backbone adaptation on small clinical datasets. The clinical implication is actionable: *for subtle diagnostic tasks, EEG-FMs should be deployed with frozen backbones*. Fine-tuning on 64 subjects overfits to idiosyncratic patterns at the cost of general spectral signatures.

CDR vs. retention ratio. The CDR–Ret.Ratio divergence illustrates why chance correction matters for robustness evaluation. CSBrain achieves the highest Ret.Ratio (0.985) but only 4th-highest CDR (0.166) because retaining near-chance performance under corruption is not meaningful robustness. The retention ratio remains a valuable brittleness diagnostic (a model with Ret.Ratio = 0.50 has genuinely collapsed), but it should not determine the ranking. The correct Axis B narrative is that *strong models degrade more in absolute terms but remain more useful*: clean performance and CDR are weakly positively correlated ($\tau \approx +0.47$).

Architectural implications. The SPA–Ret.Ratio anti-correlation ($\rho = -0.70$) suggests a concrete pretraining target: contrastive learning over channel-masked views with a spectral consistency loss,

so that representations encode spectral content regardless of which electrodes are present. The UEB-GENERAL geometry identifies the unoccupied corner (high A_m, high CDR, high C_m) as a concrete design objective for future EEG-FMs. LaBraM's θ-band collapse ($R^2 = 0.496$) is a principled limitation of VQ-VAE tokenization: the discrete codebook compresses precisely the frequency bands (θ, α, β) that carry cognitive-state information. This is not a training failure but an architectural design choice with measurable downstream consequences.

Limitations. Seven datasets do not span all EEG paradigms, and cross-paradigm generalization is not directly measured. Per-model training recipes retain each model's published hyperparameters for comparability, which may advantage or disadvantage specific models. SPA averages Welch PSD across channels, discarding spatial information; a complementary spatial metric (NAS) is reported in Supplement B. BIOT is excluded from both Axis B (STFT bypasses the dropout hook) and Axis C (STFT trivially encodes band powers), and appears in the partial UEB-GENERAL leaderboard. The D7 held-out test set contains only 8 subjects; repeated cross-validation is reported as sensitivity. Compute requirements ($\sim$3200 GPU-hours, broken down as $\approx$1200 GPU-h Axis A FBP+FT, $\approx$1900 GPU-h Axis B, $\approx$100 GPU-h kNN+SPA forward passes) limit community adoption, though a bench-lite protocol ($\sim$300 GPU-hours: 3 models $\times$ 3 datasets $\times$ 1 seed) is provided. Streaming/online inference and deployment profiling are not addressed.

8 Conclusion

UNIFORMEVALBENCH demonstrates that EEG Foundation Models should be evaluated by what their pretrained representations encode, whether those representations are grounded in neurophysiology, and whether they remain reliable under realistic signal degradation. By introducing kNN@20 (hyperparameter-free), CDR (chance-corrected robustness), SPA (prior-free spectral grounding), UEB-GENERAL (geometric composite), a novel clinical downstream task, and an adapter-based evaluation harness, we provide the community with tools to evaluate EEG-FMs on properties that matter for scientific validity and clinical deployment. Current EEG-FMs occupy distinct, non-overlapping regions of the quality–robustness–grounding space; fine-tuning can actively destroy clinically relevant representations; and no single model dominates. The three-axis geometry reveals a concrete, currently unoccupied design target—spectral fidelity from multiple electrode subsets—that should guide the next generation of EEG foundation model pretraining objectives. The epilepsy-vs-mimickers dataset, benchmark harness, and evaluation code are publicly released at `https://github.com/ANONYMOUS/UniformEvalBench`.

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A Model Pool Details

Table 9: Full model pool. Each evaluated model is integrated via a `ModelAdapter` subclass providing `encode`, `freeze_encoder`, and `unfreeze_encoder` methods. Adapters pass a conformance test suite verifying output dimensions, gradient control, and freeze/unfreeze cycle idempotency.

#	Model	Architecture	Adapter	Key Properties
M1	BIOT	Patch-freq. Transformer	`BIOTAdapter`	STFT tokenizer; fp32 required; 4 layers
M2	LaBraM	Ch.-wise NeuralTransformer	`LaBraMAdapter`	VQ-VAE tokenizer; cosine + 2-ep warmup
M3	EEGPT	ViT-style patch Transformer	`EEGPTAdapter`	Dynamic channels; 8 layers, 256-dim
M4	CBraMod	Criss-cross attn Transformer	`CBraMoDAdapter`	12 layers, 200-dim
M5	CSBrain	Region-aware Transformer	`CSBrainAdapter`	TemEmbedEEGLayer; 12 layers, 200-dim
M6	REVE	22-depth Transformer	`REVEAdapter`	GEGLU FFN; 4D Fourier PE; 512-dim
M7	MOMENT	T5-encoder	`MOMENTAdapter`	General TS-FM; 350M params

Training recipe. All models share: AdamW, head LR 5×10^{-4}, encoder LR $= 0.1\times$ head LR, 30 epochs, batch 32, grad clip 3.0, weight decay 0.01, label smoothing 0.1, bf16 AMP (BIOT: fp32). Per-model LR schedulers: EEGPT onecycle; LaBraM cosine + 2-epoch warmup; CBraMod/BIOT/CSBrain cosine; REVE reduce-on-plateau; MOMENT cosine. These match each model's published recipe.

Feature extraction. Models without a CLS token (EEGPT, CBraMod, CSBrain, REVE) use mean-pooling over patches. BIOT and LaBraM extract the CLS-equivalent token. All features are L2-normalized before probing.

B Full Axis A Results

All tables: test balanced accuracy, mean $\pm$ std over 3 seeds. kNN@20 reports mean only.

Table 10: BCIC-IV-2a (4-class motor imagery, chance = 0.25).

Model	kNN	FBP	FT	AdGap	NAG
EEGPT	**.302**	$.288 \pm .006$	$.332 \pm .013$	$+.044$	$+.062$
MOMENT	–	$.286 \pm .006$	$.364 \pm .011$	$+.078$	$+.109$
LaBraM	.251	$.278 \pm .008$	$.271 \pm .010$	$-.007$	$-.010$
CSBrain	.268	$.270 \pm .002$	$.354 \pm .005$	$+.084$	$+.115$
CBraMod	.260	$.268 \pm .002$	$.311 \pm .037$	$+.043$	$+.058$
REVE	.265	$.266 \pm .009$	$.338 \pm .010$	$+.073$	$+.099$
BIOT	.267	$.259 \pm .004$	$.334 \pm .015$	$+.075$	$+.102$

Table 11: HMC (5-class sleep staging, chance = 0.20).

Model	kNN	FBP	FT	AdGap	NAG
EEGPT	.627	$.661 \pm .001$	$.704 \pm .007$	$+.042$	$+.125$
REVE	.622	$.659 \pm .001$	$.710 \pm .004$	$+.051$	$+.150$
BIOT	.620	$.641 \pm .006$	$.681 \pm .017$	$+.040$	$+.112$
MOMENT	–	$.585 \pm .001$	$.623 \pm .003$	$+.037$	$+.090$
CSBrain	.563	$.560 \pm .001$	$.698 \pm .003$	$+.137$	$+.312$
LaBraM	.527	$.515 \pm .007$	$.642 \pm .009$	$+.127$	$+.263$
CBraMod	.485	$.509 \pm .000$	$.700 \pm .007$	$+.191$	$+.388$

Table 12: ADFTD (3-class dementia, chance = 0.33). Five of seven models show negative AdaptGap.

Model	kNN	FBP	FT	AdGap	NAG
REVE	.579	$.601 \pm .006$	$.584 \pm .022$	$-.017$	$-.044$
BIOT	.491	$.487 \pm .007$	$.384 \pm .031$	$-.103$	$-.200$
EEGPT	.441	$.432 \pm .001$	$.492 \pm .015$	$+.060$	$+.106$
MOMENT	–	$.407 \pm .004$	$.331 \pm .009$	$-.075$	$-.127$
CSBrain	.376	$.367 \pm .002$	$.510 \pm .013$	$+.144$	$+.227$
CBraMod	.364	$.359 \pm .003$	$.325 \pm .015$	$-.034$	$-.053$
LaBraM	.321	$.290 \pm .022$	$.269 \pm .024$	$-.021$	$-.029$

Table 13: Motor MV (4-class, chance $\approx$ 0.25). MOMENT excluded (OOM).

Model	kNN	FBP	FT	AdGap	NAG
EEGPT	.363	$.387 \pm .003$	$.497 \pm .006$	$+.110$	$+.179$
REVE	.291	$.312 \pm .005$	$.601 \pm .007$	$+.289$	$+.420$
BIOT	.276	$.289 \pm .002$	$.385 \pm .008$	$+.096$	$+.135$
CSBrain	.264	$.271 \pm .004$	$.549 \pm .015$	$+.278$	$+.382$
CBraMod	.260	$.265 \pm .006$	$.435 \pm .003$	$+.170$	$+.232$
LaBraM	.252	$.261 \pm .003$	$.282 \pm .006$	$+.021$	$+.028$

Table 14: Siena Scalp (binary seizure detection, chance = 0.50). MOMENT FT excluded (OOM).

Model	kNN	FBP	FT	AdGap	NAG
EEGPT	.692	$.792 \pm .007$	$.770 \pm .007$	$-.022$	$-.106$
REVE	.634	$.774 \pm .011$	$.832 \pm .009$	$+.057$	$+.254$
BIOT	.558	$.645 \pm .072$	$.677 \pm .007$	$+.032$	$+.090$
CBraMod	.554	$.619 \pm .006$	$\mathbf{.849} \pm .028$	$+.230$	$+.604$
MOMENT	–	$.540 \pm .003$	N/A	–	–
LaBraM	.500	$.500 \pm .000$	$.666 \pm .018$	$+.166$	$+.331$
CSBrain	.500	$.500 \pm .000$	$.698 \pm .023$	$+.198$	$+.397$

Table 15: Workload (binary cognitive workload, chance = 0.50).

Model	kNN	FBP	FT	AdGap	NAG
REVE	.583	$.661 \pm .035$	$\mathbf{.692} \pm .023$	$+.031$	$+.092$
BIOT	.559	$.643 \pm .028$	$.680 \pm .021$	$+.038$	$+.105$
EEGPT	.541	$.560 \pm .010$	$.670 \pm .013$	$+.110$	$+.250$
CBraMod	.505	$.500 \pm .000$	$.515 \pm .023$	$+.015$	$+.030$
CSBrain	.511	$.500 \pm .000$	$.621 \pm .062$	$+.121$	$+.242$
MOMENT	–	$.500 \pm .000$	$.553 \pm .023$	$+.053$	$+.107$
LaBraM	.502	$.499 \pm .002$	$.561 \pm .016$	$+.062$	$+.124$

C Axis B: Per-Dataset CDR and Degradation Curves

Table 16: Per-dataset CDR (chance-corrected). Bold: best per dataset. BIOT excluded (STFT bypasses dropout hook).

Model	BCIC	HMC	ADFTD	Motor	Siena	Work.	Ep-M	CDR
EEGPT	.023	**.437**	.087	**.085**	.164	**.128**	**.635**	**.223**
REVE	.012	.384	**.345**	.054	**.165**	.104	.132	.171
CSBrain	**.046**	.386	.035	.019	.000	.000	.675	.166
LaBraM	.023	.302	.002	.011	.000	.002	.546	.127
CBraMod	.024	.263	.030	.020	.000	.000	.446	.112

Table 17: Selected degradation curves (seed-averaged balanced accuracy, best-val epoch).

Model	Dataset	clean	p=0.1	p=0.25	p=0.4
EEGPT	HMC	0.665	0.669	0.490	0.490
EEGPT	Siena	0.804	0.734	0.513	0.500
EEGPT	Ep-Mim	0.880	0.891	0.855	0.707
REVE	ADFTD	0.464	**0.560**	**0.567**	**0.564**
REVE	Ep-Mim	0.832	0.691	0.507	0.500
CSBrain	Ep-Mim	0.844	0.852	0.853	0.808
LaBraM	Ep-Mim	0.760	0.760	0.779	0.781

D SPA Details

Table 18: SPA per (model, dataset), mean over 3 seeds. Bold: best per dataset.

Model	BCIC	HMC	ADFTD	Motor	Siena	Work.	Ep-M
REVE	.848	**.955**	.833	**.902**	**.922**	**.864**	**.921**
EEGPT	**.876**	.857	.825	.882	.848	.852	.883
CSBrain	.608	.880	.812	.811	.740	.604	.846
CBraMod	.597	**.913**	.825	.727	.562	.523	.834
LaBraM	.420	.713	.539	.662	.672	.377	.652

Table 19: Cross-axis Spearman ρ (5 SPA-eligible models, BIOT excluded).

Pair	ρ	Interpretation
FBP vs. SPA	$+0.90$	Spectral encoding $\Rightarrow$ probe accuracy
FBP vs. Ret.Ratio	-0.60	Stronger models more channel-dependent
Ret.Ratio vs. SPA	-0.70	Robust models encode less spectral content

E kNN Sensitivity Analysis

Table 20: k-sensitivity (max $-$ min bal. acc. across $k \in \{5, 10, 20, 50\}$). Global mean = 0.029.

Model	BCIC	HMC	ADFTD	Motor	Siena	Work.	Ep-M
EEGPT	.015	.012	.039	.013	.175	.017	.026
LaBraM	.023	.010	.029	.015	.000	.024	.066
CBraMod	.015	.027	.008	.016	.012	.006	.040
BIOT	.011	.015	.032	.020	.025	.053	.034
CSBrain	.027	.011	.025	.014	.125	.037	.035
REVE	.008	.010	.051	.019	.025	.043	.027

Table 21: kNN@20 vs FBP rank agreement (Kendall τ) per dataset. Mean $\tau = 0.519 < 0.7$: FBP is not a certified proxy for kNN@20.

Dataset	τ	p	$\tau \geq 0.7$?
BCIC-IV-2a	0.429	0.239	no
HMC	**0.867**	**0.017**	**yes**
ADFTD	**0.905**	**0.003**	**yes**
Motor MV	0.600	0.136	no
Siena	0.231	0.539	no
Workload	0.138	0.702	no
Epil-Mim	0.467	0.272	no
Mean	**0.519**		**no**

F Model-Selection Sensitivity

Table 22: FBP–FT ranking agreement under last-epoch vs. best-validation-epoch (6 core datasets; D7 best-val not yet recomputed). Both confirm Claim 1 on this subset; the 7-dataset last-epoch mean reported in Table 4 is $\tau = 0.291$. Last-epoch is primary (consistent with EEG-FM literature).

Dataset	τ (last)	τ (best-val)	Δ
BCIC-2a	+0.048	+0.333	+0.286
HMC	+0.238	+0.524	+0.286
ADFTD	+0.619	+0.714	+0.095
Motor MV	+0.467	+0.333	-0.133
Siena	+0.276	+0.447	+0.171
Workload	+0.720	+0.617	-0.103
Mean	**+0.395**	**+0.495**	**+0.100**

G Epilepsy Mimickers: Subject-Level Metadata

The epileptic cohort ($n = 50$) comprises: Generalized Epilepsy / GTCS ($n = 27$), Primary Generalized ($n = 15$), Left Temporal Focal ($n = 5$), Right Temporal Focal ($n = 3$), JME ($n = 3$), Bilateral Focal with Secondary Generalized ($n = 1$), SSPE ($n = 1$), unclassified focal ($n = 2$). Categories overlap.

The mimicker cohort ($n = 30$) comprises: PNES/PNED ($n = 4$), Neurocardiogenic Syncope ($n = 2$), Syncopal Attacks ($n = 3$), Dissociative Reaction ($n = 3$), Movement Disorders ($n = 4$), Vertigo ($n = 2$), Sleep Apnea ($n = 1$), Panic Attack ($n = 1$), Psychogenic seizure disorder ($n = 1$), Metabolic/Structural/Vascular ($n = 9$).

All recordings: 19-channel 10–20 montage (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz) at 125 Hz, linked-ear reference. EDF format, anonymized identifiers.

H Broader Impact Statement

This work introduces a benchmark and public dataset for evaluating EEG foundation models. The primary positive impact is enabling more rigorous evaluation of models proposed for clinical EEG analysis, helping identify unreliable models before clinical deployment. The epilepsy-mimickers dataset could benefit patients by accelerating differential diagnosis research.

Potential negative impacts include premature clinical deployment without prospective validation. Benchmark performance is necessary but insufficient for clinical use; all applications require validation with domain experts and regulatory oversight. The dataset is fully anonymized under IRB-approved protocols.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state five contributions (kNN@20 primary metric, chance-corrected CDR, SPA spectral grounding, novel dataset/task, UEB-General composite with adapter harness) and scope the paper to 6 EEG-FMs + 1 TS-FM reference across 7 datasets. All claims are supported by experimental results in Sections 5–6 and the pre-registered statistical framework in Section 7.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 8 discusses limitations including dataset coverage, per-model training recipe choices, SPA's spatial information loss, BIOT exclusion from SPA, D7 held-out test set size, compute requirements, absence of streaming/online evaluation, and adapter-pending models.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper does not include formal theoretical results. All metrics (kNN@20, AdaptGap, NormAdaptGain, CDR, SPA, UEB-General) are empirical quantities with clear definitions (Equations 1–6) but no associated theorems or proofs.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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   Justification: Section 3 specifies all models, adapter interfaces, and pretrained weight sources. Sections 3.2 and 4 detail dataset preprocessing, splits, and trial durations. Section 4.1 specifies the kNN@20 protocol ($k \in \{5, 10, 20, 50\}$, cosine distance, L2-normalized features). Appendix A provides per-model training recipes. All random seeds (42, 123, 7), dropout rates, chance-correction formulas, Ridge regression hyperparameters, and UEB-General composite parameters are specified. Code and data are publicly released at `https://github.com/ANONYMOUS/UniformEvalBench`.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The epilepsy-mimickers dataset (D7) is publicly released with anonymized EDF files and metadata manifest. The benchmark provides an adapter-based evaluation harness (`ModelAdapter` ABC, `DatasetAdapter`, `UEBTrainer`). Scripts to reproduce all Axis A, B, and C experiments are included. Repository: `https://github.com/ANONYMOUS/UniformEvalBench`.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

Justification: Sections 3.2 (preprocessing, splits), 4.1–4.3 (kNN@20 protocol, probing hierarchy, CDR formula, SPA implementation), and Appendix A (per-model deviations, LR schedulers, AMP precision) provide complete details. Data splits are deterministic and fully specified.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: All per-model results report mean $\pm$ standard deviation across 3 random seeds. Bootstrap 95% confidence intervals are reported for aggregate statistics. The pre-registered framework (Section 7) specifies Bonferroni correction ($\alpha = 0.05/3 = 0.0167$) and requires both $p < 0.0167$ and Cliff's $\delta > 0.33$. k-sensitivity is reported for the kNN@20 primary metric (Appendix E).

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Axis A: 240 FBP+FT + 141 kNN runs on 6$\times$H200 GPUs. Axis B: 378 runs. Overall: $\approx$4700 GPU-hours; bench-lite at $\approx$300 GPU-hours. MOMENT OOM conditions documented.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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11. **Safeguards**

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    Justification: All 6 EEG-FMs and 1 TS-FM are cited. All 6 existing datasets are cited. DINOv2 precedent for kNN evaluation is credited. The benchmark provides an independent adapter-based harness.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: D7 documented in Sections 3.2 and Appendix G. The `ModelAdapter` ABC harness is documented in Section 3.1. Dataset released with manifest CSV, anonymized EDF files, and datasheet.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants?

Answer: [N/A]

Justification: No crowdsourcing. EEG data collected during routine clinical care.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals were obtained?

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16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

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